MIS 790 Comprehensive
Exam Fall 2026

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Exam Version: MS in Information Systems

Exam Submission:

Upload your exam to Turnitin via Canvas by October 22, 2026, at 11:59pm.

Grading Exam:
Your exam is based on a 100-point score as follows:
APA Format: 20 points
Question 1: 20 points
Question 2: 20 points
Question 3: 20 points
Question 4: 20 points

APA Format (20 points)

Please visit https://apastyle.apa.org/style-grammar-guidelines/paper-format if you are unfamiliar with APA format or need a refresher.

Grading scale for each 10-point question:
Exceed expectations (9 to 10 points)
Meets expectations (6 to 8 points)
Fails to Meet expectations (0 to 5 points)

Pass/Fail Exam: 60 points or more is a passing score; 0 to 59 points is failure.

Note: Any response found to have plagiarism (based on instructor's judgement as well as Turnitin analysis) will receive a zero.

Case 1: Aztec Global Trade Disruption Intelligence

Aztec Global Freight is a multinational logistics firm that manages high-value imports into the United States from Asia and Europe, coordinating international shipments through major U.S. ports before distributing goods domestically via rail corridors, trucking fleets, and regional fulfillment hubs. Its customers include electronics manufacturers, healthcare suppliers, and industrial distributors whose operations depend on predictable transit times and stable landed costs.

In recent years, Aztec has encountered a series of disruptive events that have revealed fundamental weaknesses in how the organization anticipates and responds to risk. A hurricane at the Ports of Los Angeles and Long Beach caused vessel congestion and container backlogs that reverberated through downstream domestic routes. A major earthquake in Shanghai interrupted upstream production and export scheduling, creating delayed arrivals weeks later in U.S. distribution centers. Heightened geopolitical tensions in the Red Sea forced carriers to reroute vessels, increasing transit times and marine insurance premiums. At the same time, a CN→US tariff announcement—followed by its official effective date—generated policy uncertainty that altered procurement timing and dramatically shifted landed cost exposure. Domestically, a severe Midwest winter storm and intermittent rail network disruptions further amplified delays and missed customer commitments.

Although Aztec captures extensive shipment, routing, cost, and event data, these data streams are not structured to support forward-looking risk assessment. Disruptions are analyzed after they occur. Tariff exposure is calculated only after invoices are finalized. Domestic delays are evaluated independently from upstream international variability. Decision-making remains reactive, driven by crisis response rather than predictive insight.

More critically, leadership lacks the ability to quantify the economic tradeoffs between acting early (e.g., rerouting at tariff announcement) and waiting for confirmation (e.g., acting at the effective date). The company has no formal methodology for determining when predictive risk signals should trigger operational intervention. Managers rely on intuition rather than thresholds grounded in data.

The executive team now seeks to move beyond disruption reporting toward a structured disruption intelligence framework capable of:

- Quantifying disruption probability before service failure occurs
- Modeling the financial impact of tariff policy timing
- Evaluating cascading effects from international to domestic legs
- Comparing contingency (reactive) versus redundancy (proactive) strategies
- Establishing clear intervention thresholds tied to measurable risk scores

This initiative must strictly adhere to the five-table relational schema described in the Appendix:

- Orders
- International_Leg

- Domestic_Leg
- Event_Calendar
- Node_Lookup

Leadership does not want a reporting upgrade. They want a disciplined, analytically grounded methodology that translates predictive signals into prescriptive operational action.

You have been hired as an external consultant to design a disruption intelligence framework that integrates predictive modeling, scenario simulation, and decision thresholds into a coherent operational strategy. Your task is to develop a technology-supported approach that transforms Aztec Global Freight from reactive crisis management to structured, data-driven resilience planning.

Instructions

- Answer the four questions below.
- State any assumptions you need to make when answering. Do not seek clarification from your instructor.
- Each response should be between 500 and 1000 words. Concisely, well-supported arguments will be valued over excessive length.
- Be specific to the case study, adhere to the data model given you in the appendix. General or vague statements will not be sufficient for a passing evaluation.
- Total suggested length: ~2,000–4,000 words.

Exam Questions

1. Information Requirements (DLO 1.1)

You are required to define the critical information requirements necessary to build an effective disruption intelligence system. In your response, you must:

1.1. Explain how multi-stream event fusion (e.g., weather, conflict, tariff, and domestic operational data) should be structured and integrated.
1.2. Describe how tariff announcement dates and effective dates should be modeled distinctly and why this distinction is operationally significant.
1.3. Provide a quantitatively defensible and operational definition of disruption 'ground truth.' Clearly define the thresholds used, justify them using business logic and data characteristics, and explain how changes in threshold selection would affect system performance (e.g., false positives, missed disruptions).
1.4. Discuss how global (international leg) and domestic routing layers must be integrated within the required five-table schema.
1.5. Outline key data governance requirements, including timestamp normalization, entity standardization (ISO codes), missingness policy, and deduplication logic.

Conclude by explaining how you would validate these information requirements with stakeholders and through empirical testing.

2. Technology-Supported Architecture (DLO 1.2)
Design a feasible solution architecture aligned with the required data dictionary structure. In your response, you must:

2.1. Demonstrate how the five-table relational schema will be integrated into your system design.
2.2. Explain how predictive modeling components will be implemented and operationalized.
2.3. Describe how API endpoints (e.g., /predict, /alerts) would function within the system.
2.4. Explain how a local Streamlit dashboard would provide decision support.
2.5. Demonstrate how data integrity, governance, and relational constraints will be preserved.

Deployment may remain local; production-grade infrastructure is not required.

3. Value Measurement (DLO 1.3)
Define how the organization should measure value created by your disruption intelligence system. In your response, you must **(while giving a clear example, adhering to the given data model)**:

3.1. Compare the operational and financial impact of acting at tariff announcement versus acting at the effective date.
3.2. Explain how early disruption detection changes decision timing and performance outcomes.
3.3. Evaluate how adjustments in domestic routing strategies affect service performance.
3.4. Propose and calculate measurable KPIs, including but not limited to:
- SLA violation reduction
- Tariff exposure reduction
- Cost tradeoffs
- Operational resilience metrics

Provide clear definitions and explain how each KPI would be computed using the required schema.

4. Professional Communication (DLO 2.1–2.2)
Assume you are presenting your findings to executive leadership. In your response, you must:

4.1. Structure your written report for clarity, rigor, and executive relevance.
4.2. Provide a clear plan for responsible AI usage disclosure, including verification and human-in-the-loop principles.

Comprehensive Exam Rubric (Spring 2026)

Total: 100 Points
Passing: 60 Points

Section	DLO Alignment	Evaluation Criteria	Points
APA & Scholarly Rigor	—	Proper APA formatting, headings, citations	5
		Minimum 25 high-quality academic references	5
		Logical organization & clarity of argument	5
		Professional executive-level tone & writing quality	5
Subtotal APA			**20**
Q1: Information Requirements	DLO 1.1	Clear multi-stream event fusion structure (weather, conflict, tariff, domestic)	4
		Distinction between tariff announcement vs effective modeling	4
		Quantitatively defensible disruption definition (threshold + sensitivity logic)	4
		Correct integration of International_Leg and Domestic_Leg within five-table schema	4
		Data governance rigor (UTC normalization, ISO codes, missingness, deduplication, validation plan)	4
Subtotal Q1			**20**
Q2: Technology-Supported Architecture	DLO 1.2	Accurate integration of five-table relational schema	4
		Clear predictive model operationalization logic	4

Section	DLO Alignment	Evaluation Criteria	Points
		Coherent API logic (/predict, /alerts)	4
		Effective Streamlit dashboard design for decision support	4
		Governance, referential integrity, and access controls preserved	4
Subtotal Q2			**20**
Q3: Value Measurement	DLO 1.3	Clear comparison of acting at announcement vs effective date	4
		Explanation of early detection impact on timing & outcomes	4
		Domestic routing performance tradeoff analysis	4
		Well-defined and calculated KPIs (SLA, tariff exposure, cost tradeoffs, resilience)	4
		Explicit demonstration of how KPIs are computed using required schema	4
Subtotal Q3			**20**
Q4: Professional Communication	DLO 2.1–2.2	Executive-ready written report structure	5
		Clear oral presentation strategy (flow, visuals, persuasion)	5
		Tradeoff framing (cost vs resilience vs operational feasibility)	5
		Responsible AI disclosure (verification + human-in-the-loop)	5

Section	DLO Alignment	Evaluation Criteria	Points
Subtotal Q4			**20**
TOTAL			**100**